VISHNU DHARSAN A S

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SUMMARY

Software Engineer with 1+ year of experience building enterprise AI-powered applications using Python, FastAPI, LangChain, Ollama, PostgreSQL, MongoDB, and Retrieval-Augmented Generation (RAG). Experienced in designing scalable backend REST APIs, semantic search pipelines, NLP, prompt engineering, and AI assistants for industrial automation. Seeking Software Engineer, AI Engineer, or GenAI Engineer roles.

TECHNICAL SKILLS

Programming Languages: Python, C, SQL, Java, JavaScript, HTML, CSS

AI / ML / NLP: Retrieval-Augmented Generation (RAG), LangChain, Hugging Face Transformers, OpenAI API, Ollama, Embeddings, Semantic Search, Vector Databases (FAISS, ChromaDB), MCP, TensorFlow, Keras, Scikit-learn, LightGBM, NumPy, Pandas

Backend & APIs: FastAPI, Flask, REST APIs, Node.js, Next.js, React

Databases: PostgreSQL, MongoDB, MySQL, ChromaDB

Cloud, DevOps & Tools: Linux, Docker, Podman, Git, GitHub, Google Cloud Platform (GCP), Jupyter Notebook, VS Code

Data Analytics & Visualization: Power BI, MS Excel, Matplotlib, Seaborn, Plotly

Concepts: Prompt Engineering, Context Management, Document Ingestion, Data Preparation & Cleansing, AI Agents, System Design, Data Structures & Algorithms, Object-Oriented Programming (OOP)

EXPERIENCE

ASM Technologies (Software Engineer) | Aug 2025 – Present

- Designed and developed an enterprise AI assistant for industrial failure analysis by building an end-to-end Retrieval-Augmented Generation (RAG) solution for troubleshooting, log analysis, and failure analysis (FA) report generation.
- Built scalable document ingestion, data preparation, and semantic retrieval pipelines using Ollama LLMs, ChromaDB, and LangChain, indexing 1,000+ technical documents and logs to enable accurate, context-aware responses from technical documentation and domain knowledge.
- Developed REST APIs using FastAPI and Flask, integrating PostgreSQL and MongoDB for efficient data management, chat history, and application workflows; reduced average API response latency by ~35% through async endpoints, query optimization, and caching.
- Containerized and deployed AI services using Podman/Docker for reproducible, portable deployments.
- Implemented prompt engineering, retrieval optimization, and context management to enhance response quality and reduce hallucinations by ~40%, enabling log analysis, root cause identification, and citation-backed FA report generation for industrial troubleshooting, reducing manual report preparation time by ~60%.

AES Technologies (Project Intern) | Oct 2023 – May 2024

HRIQ – HRMS System

- Full-stack development of an HRMS project for employee management, with ownership of the Reimbursement feature.
- Analyzed employee data to identify trends and improve reimbursement processing efficiency using SQL and .NET.
- Collaborated with a cross-functional team of 6, using jQuery, C#, and SQL Server to deliver a scalable and modular HR platform, decreasing feature deployment time by 20%.

PROJECTS

Automotive Failure Detection and Recovery Recommendation

- Created a hybrid LLM + rules-based system that classified automotive failures with ~90% accuracy across 4 OEM datasets, generated recovery recommendations, and built a web UI.
- Compared outputs of different LLMs to select the best-suited model for performance.
- *Tech:* Python, LLMs (GPT-3, Llama 3, Bard/Gemini, Claude, Falcon), rule-based classification, React, Node.js.

Bearing Fault Classification (Paper Published)

- Developed a deep learning model to predict faults and remaining lifetime of bearings used in machines, including data preprocessing and feature preparation.
- *Tech:* Python, CNN, ANN, TensorFlow, SVM, LightGBM.

HR Data Analytics & Marketing Analysis (Python, SQL Server, Power BI, Pandas, NumPy)

- Analyzed HR data to derive actionable insights on employee performance, retention, and workforce optimization, using data analysis, warehousing, and visualization.
- Analyzed shop data to understand customer behavior and helped create a better business strategy.

EDUCATION

PSG College of Engineering, Coimbatore, TN

Bachelor of Engineering – Computer Science and Engineering | CGPA: 8.0 | Nov 2021 – Apr 2025

CS Academy, Erode, TN

Class XII – 94% (May 2021) | Class X – 92% (May 2019)

ACHIEVEMENTS & ACTIVITIES

- Represented ASM Technologies at CII INNOVERGE 2025.
- Delivered a technical talk: "Building RAG Chatbot using NVIDIA NIM".
- Secretary, Entrepreneur's Club (event organisation, speaker sessions, overall management); Photography and Publicity Head, CSE Association; Class Representative (5th semester).